Forecasting Bike Rental Demand

UE24CS352A Machine Learning - Mini-Project | Team: Greeshma K (PES1UG24AM368), Saurav S (PES1UG24AM398) | Problem: Forecasting Bike Rental Demand | Repo: https://github.com/sauravs2006-max/Forecasting-Bike-Rental-Demand-ML-MINIPROJ (private, shared with faculty/TAs)

1. Problem Statement

Predict the total number of bikes rented **per hour** in Washington D.C.'s Capital Bikeshare system from date/time and weather information (Kaggle *Bike Sharing Demand*; reference: Du, He & Zhechev, Stanford CS229, 2014). The metric is RMSLE = sqrt(mean((log(p+1) - log(a+1))^2)), which penalises relative rather than absolute error.

2. Dataset

10,886 hourly training rows from 2011-01-01 to 2012-12-19 (days 1-19 of each month) and 6,493 test rows (later days of each month, no labels). Features: datetime, season, holiday, workingday, weather (1-4), temp, atemp, humidity, windspeed. Targets: casual, registered, count (= casual + registered; **count** is predicted). Missing values: 0. **EDA findings:** weekday demand peaks around hours 17 and 18, weekend demand peaks at hour 13; count is right-skewed (mean 192, median 145, skew 1.24), so models are trained on log(1+count); correlation of count with atemp is 0.39. Mean hourly demand was 1.65x higher in 2012 than in 2011.

3. Approach

Feature engineering. Hour and year are extracted from datetime; six optional ideas from the paper (month instead of season, day-of-week, drop temp, drop holiday, peak-hour flags, discretised weather) were searched over all 64 subsets with a fast decision tree, scoring each by the mean of 5-fold CV and temporal hold-out RMSLE. Best subset: **month instead of season, day-of-week, holiday dropped, weekday/weekend peak-hour flags, discretised weather variables**. **Models.** Linear regression, Poisson GLM, PCR, SVR (RBF), gradient boosting (HistGradientBoosting), Random Forest, Decision Tree, and a log-space stack (tree + RF + GBM -> linear meta-model). Regressors are trained on log(1+count) so squared error equals the RMSLE objective; hyper-parameters come from grid search with K-fold CV. **Validation.** Random K-fold leaks neighbouring hours, so we also report a temporal hold-out (train: days < 15, validate: days 15-19) that mimics the real split (test = days 20+). The winner is refit on all training data.

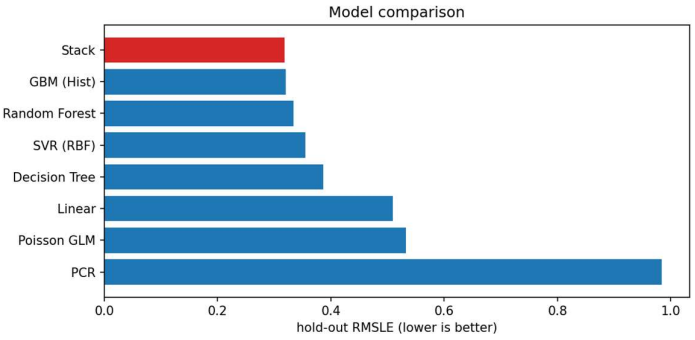
4. Implementation Overview

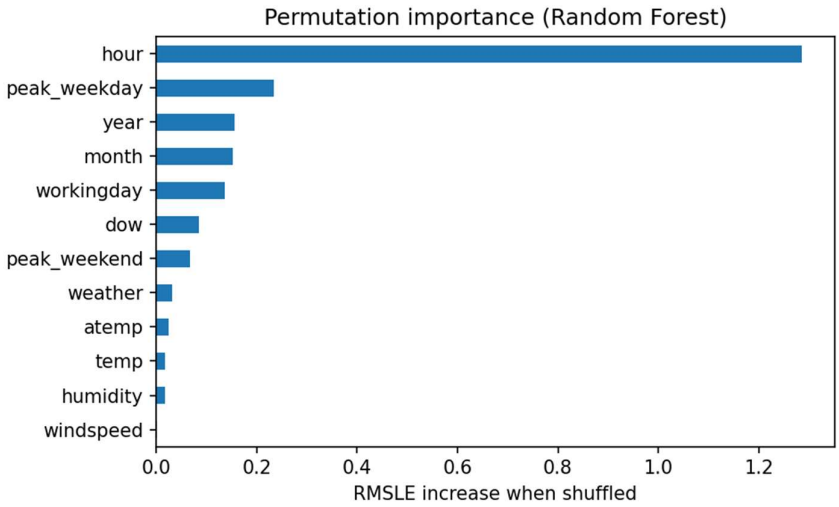
Python 3 / scikit-learn. `features.py` (feature flags) -> `models.py` (model zoo, LogStack) -> `ablation.py` / `train.py` (search, tuning, comparison, saving) -> `predict.py` (Kaggle-format submission) and `demo.py` (single-hour prediction). Everything is reproducible with `bash run_all.sh`; seeds are fixed.

5. Results

Model	CV RMSLE	Hold-out RMSLE
Stack	-	0.3182
GBM (Hist)	0.2728	0.3204
Random Forest	0.2988	0.3339
SVR (RBF)	0.3084	0.3555
Decision Tree	0.3582	0.3869
Linear	0.5115	0.5093
Poisson GLM	0.5556	0.5322
PCR	0.9443	0.9838

Rows are sorted by hold-out RMSLE (lower is better); the shaded row is the selected model.





6. Conclusions

- The best model, Stack, reached a hold-out RMSLE of 0.3182 (CV RMSLE: n/a for the stack).
- Tree-based/ensemble models (Stack: 0.3182) beat the best linear-family model (Linear: 0.5093), consistent with a non-linear dependence of demand on hour, working day and weather.
- Most important features (permutation importance, Random Forest): hour, peak_weekday, year.
- The 64-subset feature search lowered the mean (CV, hold-out) RMSLE from 0.4151 (no optional features) to 0.3743.
- Limitations: the hold-out is days 15-19 of each month, a proxy for the real test split (days 20+); no lag/rolling demand features (this is not a true time-series model); casual and registered riders are not modelled separately; one random seed and a limited grid.
- Future work: lag and rolling features, cyclic hour encoding, separate casual/registered models, per-model feature sets, wider hyper-parameter search.